

Review of Traps from Last Semester (1): Number and Quantity

Comprehensive review of trap in the first half of the semester · 70 minutes · 1-to-3 Online Lesson

Review scope: Lesson 1-2 (Multiple digits and estimation) + Lesson 7-10 (Fraction operations) + Lesson 12-13 (Decimal operations)

Core traps: □ T1 Multi-digit traps + T2 Sequence of operations + T9 Fractional operations ● SSPA High frequency

SSPA related: ● High frequency Approximately 30-35% of the test paper is scored, and the number and quantity are the largest proportion of the entire paper

Prerequisite knowledge: Hall 1-2 (multiple digits/estimate) · Hall 7-10 (score) · Class 12-13 (Decimals)

Objectives of this class: ❶ Review of high-frequency traps in multi-digit and estimation ❷ Comprehensive review of fraction traps ❸ Comprehensive review of decimal traps

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Express 3,056,000 in units of "ten thousand".	Basic	
2	Estimate: $497 + 302 \approx ?$ (rounded to the nearest hundred)	Basic	
3	Calculate $\text{LCM}(6, 8) = ?$	Basic	
4	Convert 0.35 to its simplest fraction.	Advanced	
5	Compare $\frac{5}{8}$ and 0.6, which one is larger? Write down the judgment process.	Advanced	

II.Core Knowledge + trapreview

Knowledge Point 1: Multi-digit + Estimation Trap Review ● T1 Trap

T1 multi-digit high-frequency trap review:

- ① **Place value confusion** |||SEP|||: The "3" of 1234567 is in the "ten thousand" place instead of the "hundred thousand place" - the fifth digit from the right is the ten thousand place
Unit conversion error |||SEP|||: $10,000 = 10,000$, $3.056 \text{ million} = 3,056,000$ (the decimal point is shifted to the right by 4 places)
- ② **The estimation digits are not uniform** |||SEP|||: The addition and subtraction estimations use the same digit value, and the multiplication and division estimations use compatible numbers (compatible numbers)
Wrong rounding direction |||SEP|||: Look at the next digit, carry if ≥ 5 , round if < 5
- ③ **Approximate value symbol** |||SEP|||: Use " $\approx$ " for estimation, not " $=$ " (SSPA points deduction!): The same place value is used for addition and subtraction estimation, and compatible numbers are used for multiplication and division estimation.
- ④ **Wrong rounding direction**: Look at the next digit, ≥ 5 is carried, < 5 is discarded
- ⑤ **approximation symbol**: Use " $\approx$ " for estimation, not " $=$ " (SSPA points deduction!)

□ **Trap detonation example question 1 (T1 multi-digit - place value trap, SSPA required test every year)**

In the number 4,567,890, where is "6"? What is its value?

✗ **Common mistakes (65% students)**

"6" in "one hundred thousand"

Counting from the left - wrong! Place values are counted from the right. The correct answer is: 5th digit from the right = 10,000 digits

✓ **Correct answer**

"6" in "ten thousand" → value =
60,000

Counting from the right: 0=pieces, 9=tens, 8=hundreds, 7=thousands, 6=ten thousand, 5=hundreds of thousands, 4=millions

Example 2 - Estimation Trap (T1 - Inconsistent positioning)

Estimate: $486 + 319 \approx ?$ (rounded to the nearest hundred)

✗ Common mistakes

$$500 + 319 = 819 \approx 800$$

One has taken the hundreds place (500), the other has not taken it (319) - both must have the same place value!

✓ Correct approach

$$486 \approx 500, 319 \approx 300 \rightarrow 500 + 300 = 800$$

First take the divisor of each (take the hundreds place together), and then calculate

Tip: "Count the place value from the right, estimate the place first, add and subtract with the same accuracy, and multiply and divide compatible numbers."

⚠ T1 The most frequent trap: place value counting from the left vs. counting from the right - always start counting from the right (ones digit)!

⚠ T1 second high frequency: use "=" instead of "≈" for estimation, 1 point will be deducted for the test!

Knowledge Point 1 Synchronization practice

#	Question	Difficulty	Working Space
Example 1	In the number 2,345,678, where is "4"? What is its value?		
Example 2	Estimate: $2,130 + 4,870 \approx ?$ (rounded to the nearest thousand)		

Knowledge Point 2: General Review of Fraction Traps T9 Trap

A comprehensive review of T9 high-frequency traps in fraction operations:

- ① **Direct addition and subtraction of different denominators** : $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$ (numerator + numerator, denominator + denominator) - the most common mistakes!
- ② **The numerator forgot to expand simultaneously after the common denominator** |||SEP|||: **Only the denominator is changed, the numerator remains unchanged** The answer is not reduced |||SEP|||: The answer to the split test must be the simplest fraction
- ③ **The improper fraction is not converted into a mixed fraction** |||SEP|||: **When the numerator > the denominator, it must be converted into a mixed number** LCM calculation error |||SEP|||: Common multiples are used but not the "minimum"
- ④ **Integers and fractions are out of touch when operating with mixed fractions** |||SEP|||: **It is recommended to use the "Conversion to Improper Fractions Method"**: When numerator > denominator, it must be converted into a mixed number
- ⑤ **LCM calculation error**: A common multiple is used but it is not the "minimum" one
- ⑥ **There is a disconnect between integers and fractions when operating with mixed numbers**: It is recommended to use the "Conversion to Improper Fractions Method"

□ Trap detonation example question 3 (T9 - direct addition of different denominators, the most serious penalty for SSPA subtests)

calculate: $\frac{2}{3} + \frac{1}{4} = ?$

✗ **Fatal errors (source of 80% of SSPA points lost)**

$$\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$$

Add numerator to numerator ($2+1=3$), add denominator to denominator ($3+4=7$)—all wrong!

✓ **Correct solution (three-step method)**

$$\text{LCM}(3,4)=12 \rightarrow \frac{8}{12} + \frac{3}{12} = \frac{11}{12}$$

Simultaneous expansion of the numerator: $2 \times 4=8$, $1 \times 3=3 \rightarrow$
Add the numerators without changing the denominator

Example 4 - Mixed fraction arithmetic trap + unreduced answer

Calculate: $1\frac{2}{6} + 2\frac{1}{3} = ?$

✗ **Common mistakes**

$$1\frac{2}{6} + 2\frac{1}{3} = 3\frac{3}{9}$$

Add integers and add fractions directly (and add different denominators directly!)

✓ **Correct solution (conversion to improper fraction method)**

$$= \frac{8}{6} + \frac{7}{3} = \frac{8}{6} + \frac{14}{6} = \frac{22}{6} = 3\frac{2}{3}$$

Conversion: $126=86$, $213=73 \rightarrow$ common denominator $\rightarrow$
calculation $\rightarrow$ reduction

🧠 **Tip: "There are three things you can't do when adding fractions: you can't add if the denominators are different, you can't add if you don't agree on the numerators, and you can't get the answer if you agree on it."**

⚠ **T9 Highest Frequency: Direct addition and subtraction of different denominators without adding common fractions - > 50% of candidates make this mistake every year in the score test paper!**

⚠ **T9 Second High Frequency: The answer is not reduced - 612 is not the final answer, it is reduced to 12!**

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
Example 3	$\frac{3}{5} + \frac{1}{2} = ?$		
Example 4	$2\frac{1}{4} + 1\frac{2}{3} = ?$		

Knowledge Point 3: Review of Decimal Traps ● T9 Traps

Comprehensive review of high-frequency traps in decimal operations:

- ① **Decimal point alignment errors** |||SEP|||: **The decimal points are not aligned when adding and subtracting - tenths to tenths, hundredths to hundredths**Error in the number of decimal places in multiplication |||SEP|||: The decimal place of the product = the sum of the decimal places of each factor ($0.3 \times 0.4 = 0.12$)
- ② **The decimal point shift is inconsistent in division** |||SEP|||: **The divisor and the dividend must be shifted synchronously (multiplied by 10/100/1000)**Decimal comparison only looks at the number but not the place value |||SEP|||: $0.8 > 0.75$ (not $0.75 > 0.8$! The latter does not mean that the latter has more decimal places)
- ③ **Wrong denominator when converting fractions to decimals**(two decimal places $\rightarrow$ denominator 100), not
- ④ **Forgot to add zeros when multiplying/dividing decimals**: $0.8 > 0.75$ (not $0.75 > 0.8$! More decimal places in the latter does not mean large)
- ⑤ **Wrong denominator when converting fractions to decimals** : $0.35 = \frac{35}{100}$ (two decimal places $\rightarrow$ denominator 100), no $\frac{35}{10}$
- ⑥ **Forgot to add zeros when multiplying/dividing decimals** : $0.4 \times 0.5 = 0.20 = 0.2$

❑ **Trap detonation example 5 (T9 - decimal multiplication digit trap)**

Calculation: $0.3 \times 0.4 = ?$

✗ **Common mistakes (45% students)**

$$0.3 \times 0.4 = 0.12$$

This is right! The trap is: $0.3 \times 0.4 \neq 0.12$? wrong! The real trap is for students to write $0.3 \times 0.4 = 1.2$ (forgetting two decimal places)

✓ **Correct rules**

$$3 \times 4 = 12 \rightarrow 0.3 \times 0.4 = \mathbf{0.12}$$

Factors each have 1 decimal place, 2 in total $\rightarrow$ product 2 digits from the right plus decimal point: $12 \rightarrow 0.12$

Example 6 - Decimal point shift trap in decimal division

Calculation: $3.6 \div 0.12 = ?$

✗ **Common mistakes**

$$3.6 \div 0.12 = 0.3$$

Just change 3.6 to 36, forget about 0.12 and shift it synchronously!

✓ **Correct approach**

$$\text{Divisor} \times 100 \rightarrow 0.12 \times 100 = 12$$

$$\text{Dividend} \times 100 \rightarrow 3.6 \times 100 = 360$$

$$360 \div 12 = \mathbf{30}$$

Shift both sides simultaneously: the divisor has 2 decimal places $\rightarrow$ multiply both sides by 100

 **Tip: "Align the points for addition and subtraction of decimals, add and bury the number of multiplication digits, and move both sides of division simultaneously. It is better not to move only one side."**

 **T9 The most common trap with decimals: Forgetting to calculate the decimal places in multiplication - $0.3 \times 0.5 = 0.15$ (not 1.5! Not 0.015!)**

 **T2 order of operations trap: when decimals are mixed, multiply and divide first, then add and subtract, parentheses have the highest priority!**

III. Lesson Layered Synchronization Practice

Basic level (total 7 questions, all must do - question basic skills in each lesson)

#	Question	Difficulty	Working Space
6	In the number 9,087,654, where is "0"? What is its value?		
7	Estimate: $3,510 - 1,280 \approx ?$ (rounded to the nearest hundred)		
8	$\frac{1}{6} + \frac{1}{3} = ?$		
9	$\frac{3}{4} + \frac{1}{8} = ?$		
10	Calculation: $0.7 + 0.35 = ?$		
11	Calculation: $0.6 \times 0.5 = ?$		
12	Convert 0.08 to its simplest fraction.		

Advanced layer (total 7 questions,  choose do — trap mixed question)

#	Question	Difficulty	Working Space
13	Write 4.056 million in Arabic numerals.		
14	Estimate: $49 \times 31 \approx ?$ (Using consistent number method: 50×30)		
15	$\frac{5}{8} + \frac{1}{6} = ?$ (Answer must be simplified)		
16	$2\frac{2}{5} + 1\frac{3}{10} = ?$		
17	Calculation: $1.25 \div 0.5 = ?$		
18	Calculation: $0.3 \times 0.2 \times 0.5 = ?$		
19	Compare $\frac{3}{8}$, 0.4 , $\frac{2}{5}$ and arrange from largest to smallest.		

 challenge layer (total 6 questions,  choose do — Cross-Topiccomprehensive)

#	Question	Difficulty	Working Space
20	$1\frac{3}{4} + 2\frac{5}{6} - 1\frac{1}{2} = ?$		
21	Calculation: $0.75 \times \frac{4}{5} = ?$ (Answer in decimals)		
22	A ribbon is $5\frac{3}{8}$ meters long. 2.5 meters are used. How many meters are left? (Answer in fractions)		

23	Estimate: $596 \times 198 \approx ?$ (Use consistent number method, round to the nearest hundred)		
24	$\frac{3}{4} + 0.25 + \frac{1}{8} - 0.375 = ?$		
25	The sum of the three fractions is 2: the first is $\frac{3}{4}$ and the second has more $\frac{1}{6}$ than the first. What is the third score?		

IV. application questions special topic training (total 6 questions)

Life application question (all must do, column formula → calculate → answer sentence)

#	Question	Difficulty	Working Space
26	Hong Kong has a population of approximately 7,500,000. Express this number in units of "ten thousand".		
27	Buy a book for \$48.5 and a pen for \$12.8. How much is the estimated total? (Round to the nearest ten dollars)		
28	The cake shop sold $\frac{5}{8}$ cakes throughout the day and $\frac{1}{4}$ cakes in the morning. How many were sold in the afternoon?		
29	Xiao Ming has 0.75 kilograms of candy, Xiao Qiang has 0.5 kilograms, and Xiao Mei has $\frac{3}{5}$ kilograms. How many kilograms of candy do the three of them have in total? (Answer in decimals)		
30	There are 35 students in the classroom. Among them $\frac{2}{5}$ are boys, and the rest are girls. What fraction of the number of girls are there as compared to boys?		
31	Restaurant A has 120 seats, and restaurant B has 1.5 times as many seats as restaurant A. How many seats are there in the two restaurants?		

V. Ultimate challenge area (total 3 question)

#	Question	Difficulty	Working Space
 1	The water tank contains $3\frac{3}{4}$ liters of water. The water in pipe A is 0.5 liters per minute, and the water in pipe B is $\frac{1}{4}$ liters per minute. How many liters of water are in the tank after the two tubes are opened for 5 minutes?		
 2	The $\frac{2}{5}$ of a number is equal to 0.6. What is the $\frac{3}{4}$ of this number?		

3	A, B, and C share a sum of money. Point A gets SEP , point B gets 0.35, and point C gets \$240. How much is the total amount of money? $\frac{2}{5}$, B gets 0.35, and C gets \$240. How much is the total amount of money?		
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VI. Class after homework

Basic must-do questions (total 5 questions)

#	Question	Difficulty	Working Space
H1	In the number 5,432,109, where is "4"?		
H2	Estimate: $2,870 + 4,350 \approx ?$ (rounded to the nearest hundred)		
H3	$\frac{2}{5} + \frac{3}{10} = ?$		
H4	Calculation: $0.8 \times 0.25 = ?$		
H5	Compare $\frac{7}{10}$ and 0.65, which one is larger?		

Advanced choose do question (total 3 questions, choose do)

#	Question	Difficulty	Working Space
H6	$1\frac{2}{3} + \frac{5}{6} - 0.5 = ?$		
H7	Calculation: $2.4 \div 0.06 = ?$		
H8	Garden area 120 m ² . $\frac{1}{3}$ Plant roses, 0.25 species chrysanthemums. How many square meters does the flower planting area occupy in total?		

VII. The Lesson core common errors summary

Review of Learning Objectives - After completing this lesson you should be able to:

- ☐ Identify all trap types in our hall
 ☐ Solve basic questions independently (100% correct)
 ☐ Challenge Advanced questions (80%+ correct)
 ☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	The place value is counted from the left SEP : The number 1234567 says "3" is in the 100,000th place: The number 1234567 says "3" is in the 100,000th place	Place values are counted from the right (ones digit): tens, hundreds, thousands, hundreds, thousands, millions
2	Estimates are inconsistent : $500 + 319 \approx 500 + 300$	All numbers are calculated with the same accuracy.

3	Direct addition and subtraction of fractions with different denominators (T9)	Different denominators → must be connected first (find LCM)
4	Fractional answers are not reduced (T9)	The last step is to check the common factors and reduce to the simplest fraction.
5	Wrong number of decimal places to multiply : $0.3 \times 0.4 = 1.2$ (T9)	Decimal places of product = sum of decimal places of each factor
6	Decimal division only moves one side (T9)	The divisor and dividend are shifted simultaneously (multiplied by 10/100/1000)
7	Mixed fractions and decimals are not in a unified format (T2)	Unify the format first (convert all fractions or all decimals) and then calculate